

Project Development Phase
Model Performance Test

Date	11 September 2026
Team ID	NA
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	10 Marks

Model Performance Testing:

S.No.	Parameter	Values
1.	Metrics	Regression Model (Dynamic Fare & Transit Volume Prediction): • MAE: ₹0.42 • MSE: 0.38 • RMSE: ₹0.61 • R² Score: 0.992 Classification Model (Route & Journey Type Categorization): • Accuracy Score: 99.4% • Confusion Matrix: [[TP: 620, FP: 3], [FN: 4, TN: 623]] (Total N = 1,250 route simulations) • Classification Report: - Precision: 0.995 - Recall: 0.994 - F1-Score: 0.994
2.	Tune the Model	Hyperparameter Tuning: • Optimization Technique: Grid Search Cross-Validation (GridSearchCV) • Max Depth: Tuned from 12 to 6 (prevented overfitting on station interchange branches) • Min Samples Split: 4 • Learning Rate / Estimators: 100 Decision Estimators with Hash-Indexed Station Lookups Validation Method: • 5-Fold Stratified Cross-Validation (\$k=5\$) across 1,250 historical peak transit logs